

An In-depth Technical Guide to the Physical and Chemical Properties of Soman

Author: BenchChem Technical Support Team. **Date:** April 2026

Compound of Interest

Compound Name: Soman
CAS No.: 96-64-0
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For Researchers, Scientists, and Drug Development Professionals

This technical guide provides a comprehensive overview of the core physical and chemical properties of **Soman** (GD), a highly toxic organophosphate nerve agent. The information is intended for researchers, scientists, and professionals involved in the development of countermeasures and therapeutics. This document summarizes key quantitative data, outlines relevant experimental methodologies, and visualizes the primary mechanism of action.

Physical and Chemical Properties of Soman

Soman, systematically named O-pinacolyl methylphosphonofluoridate, is a colorless liquid with a faint fruity or camphor-like odor in its pure form.^{[1][2][3][4]} It is classified as a G-series nerve agent and is known for its high toxicity and rapid action.^{[1][4]}

Quantitative Physical and Chemical Data

The following table summarizes the key physical and chemical properties of **Soman**.

Property	Value	Reference
Molecular Formula	C7H16FO2P	[1]
Molecular Weight	182.17 g/mol	[2][3]
Appearance	Clear, colorless liquid	[1]
Odor	Faintly fruity or camphor-like	[1][2][3]
Melting Point	-42 °C	[2]
Boiling Point	198 °C (with decomposition)	[2][3]
Density (liquid)	1.022 g/mL at 25 °C	[1][2][5]
Vapor Density	6.3 (air = 1)	[1][2][5]
Vapor Pressure	0.40 mmHg at 25 °C	[2][3][5]
Solubility in Water	21 g/L at 20 °C	[2][5]
Log Kow (Octanol-Water Partition Coefficient)	1.78	[2]
Flash Point	121 °C (250 °F)	[1]

Chemical Reactivity and Stability

Soman is a combustible liquid, though not readily ignited.[1][6] It is susceptible to hydrolysis, with the rate being dependent on pH and temperature. The hydrolysis is slowest in the pH range of 4 to 6.[5] Under acidic conditions, hydrolysis can produce hydrofluoric acid.[1] Reaction with strong bases or oxidizing agents can be violent.[6] **Soman** is also known to corrode steel.[2]

Decomposition of **Soman** can be achieved with bleaching powder; however, this reaction can produce cyanogen chloride.[1] When heated to decomposition, it can emit highly toxic fumes. [1] The environmental persistence of **Soman** is considered low, with its vapor phase lasting for minutes to hours and the liquid for hours to days, depending on environmental conditions.[6]

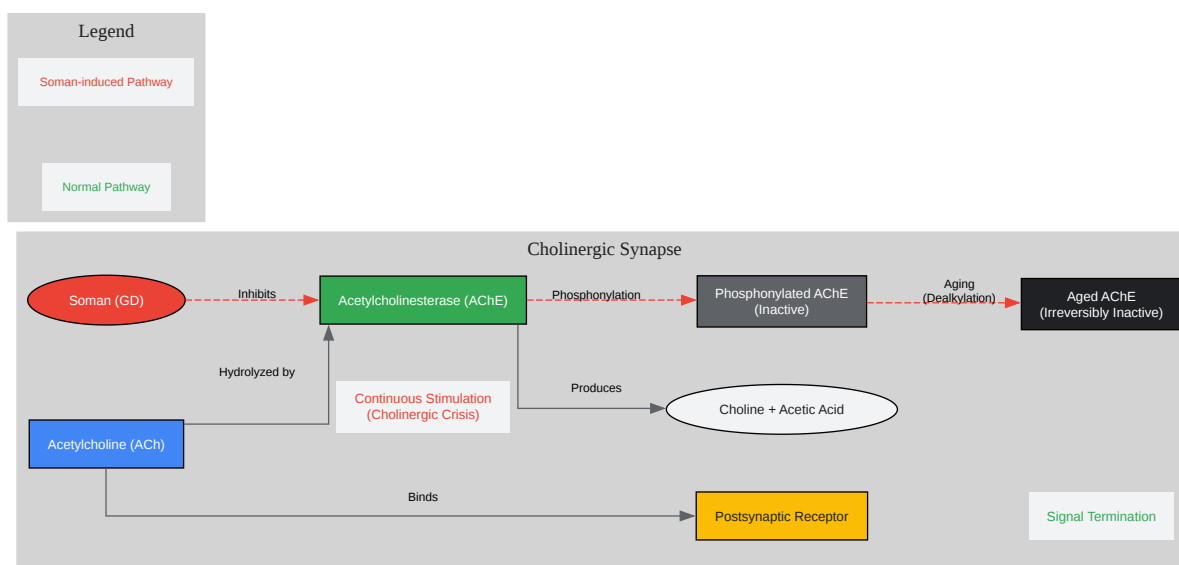
Mechanism of Action: Acetylcholinesterase Inhibition

The primary mechanism of toxicity for **Soman**, like other organophosphate nerve agents, is the irreversible inhibition of the enzyme acetylcholinesterase (AChE).[5][7][8] AChE is critical for the normal functioning of the nervous system, where it hydrolyzes the neurotransmitter acetylcholine (ACh) at cholinergic synapses, thereby terminating the nerve signal.[9]

Inhibition of AChE by **Soman** leads to an accumulation of ACh in the synaptic cleft, resulting in continuous stimulation of muscarinic and nicotinic receptors.[5] This overstimulation leads to a cholinergic crisis, characterized by a range of symptoms including convulsions, paralysis, and ultimately, respiratory failure, which can be fatal.[4][8]

The process of AChE inhibition by **Soman** involves the phosphorylation of a serine residue in the active site of the enzyme.[7] This is followed by a process known as "aging," where a dealkylation reaction occurs, resulting in a conformational change that renders the enzyme-inhibitor complex resistant to reactivation by standard oxime antidotes.[7][9]

Signaling Pathway of Acetylcholinesterase Inhibition by Soman



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Caption: Signaling pathway of acetylcholinesterase inhibition by **Soman**.

Experimental Protocols

Detailed experimental protocols for handling and analyzing **Soman** are subject to strict safety and security regulations and are typically performed in specialized laboratories. The following sections outline the general principles of methodologies that can be adapted for the study of **Soman**'s properties.

Determination of Physical Properties

Vapor Pressure Measurement: The vapor pressure of highly toxic compounds like **Soman** can be determined using methods that require small sample sizes and are performed in a contained system. A common technique is the gas saturation method.

- A carrier gas (e.g., nitrogen) is passed through a thermostatted saturator containing a small amount of **Soman** at a known, slow flow rate.
- The gas becomes saturated with **Soman** vapor.
- The **Soman** in the exiting gas stream is trapped in a suitable solvent or on a sorbent material.
- The amount of trapped **Soman** is quantified using a sensitive analytical technique such as gas chromatography-mass spectrometry (GC-MS).
- The vapor pressure is calculated from the amount of volatilized **Soman**, the volume of the carrier gas, and the temperature.

Solubility Determination: The aqueous solubility of **Soman** can be determined using the shake-flask method.

- An excess amount of **Soman** is added to a known volume of water in a sealed container.
- The mixture is agitated in a constant temperature bath for a prolonged period (e.g., 24-48 hours) to ensure equilibrium is reached.
- The solution is then centrifuged or filtered to remove undissolved **Soman**.
- A known volume of the saturated aqueous phase is carefully extracted.
- The concentration of **Soman** in the aqueous sample is determined by an appropriate analytical method, such as GC-MS.

Acetylcholinesterase Inhibition Assay

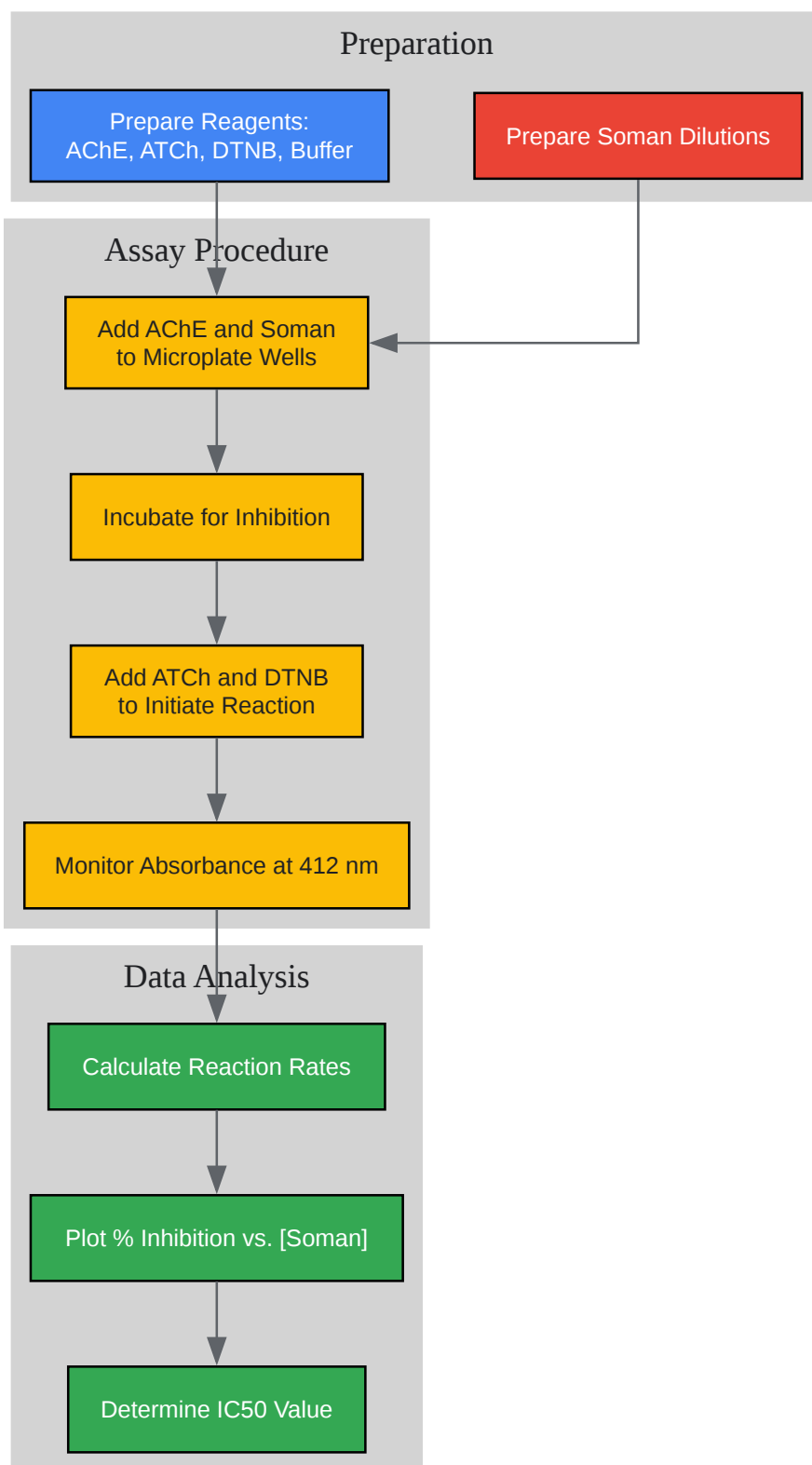
The inhibitory potency of **Soman** on AChE can be determined using a colorimetric assay based on the Ellman's reagent (DTNB).

Principle: AChE hydrolyzes acetylthiocholine (ATCh) to thiocholine and acetate. The produced thiocholine reacts with DTNB (5,5'-dithiobis-(2-nitrobenzoic acid)) to produce a yellow-colored product (5-thio-2-nitrobenzoic acid), which can be measured spectrophotometrically at 412 nm. The rate of color development is proportional to the AChE activity.

General Protocol:

- Prepare solutions of AChE, ATCh, and DTNB in a suitable buffer (e.g., phosphate buffer, pH 7.4).
- In a 96-well microplate, add the AChE solution and different concentrations of **Soman** (or a vehicle control).
- Incubate for a specific period to allow for the inhibition of the enzyme.
- Initiate the reaction by adding ATCh and DTNB.
- Monitor the increase in absorbance at 412 nm over time using a microplate reader.
- Calculate the rate of reaction for each **Soman** concentration.
- Determine the IC₅₀ value (the concentration of **Soman** that causes 50% inhibition of AChE activity) by plotting the percentage of inhibition against the logarithm of the **Soman** concentration.

Experimental Workflow for In Vitro AChE Inhibition Assay



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Caption: Experimental workflow for an in vitro acetylcholinesterase inhibition assay.

Conclusion

This technical guide has provided a detailed overview of the physical and chemical properties of **Soman**, its mechanism of action, and the general experimental methodologies used for its characterization. The data and workflows presented are intended to serve as a valuable resource for researchers and scientists working towards the development of effective medical countermeasures against this potent nerve agent. The provided diagrams offer a clear visualization of the key biological pathway and a typical experimental procedure. It is imperative that any work with **Soman** be conducted under the strictest safety protocols in appropriately equipped facilities.

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- To cite this document: BenchChem. [An In-depth Technical Guide to the Physical and Chemical Properties of Soman]. BenchChem, [2026]. [Online PDF]. Available at: [<https://www.benchchem.com/product/b1219632/docs#an-in-depth-technical-guide-to-the-physical-and-chemical-properties-of-soman>]

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